Supplemental Table 1: Statistical analyses of the specific outcomes in each study

Study	Study characteristics	Intervention	Results	P-value ()/CI{}
Alsaleem et al., 2020 [11]	**Setting:** USA, Buffalo, NY. Neonatal intensive care units (NICUs) at Women and Children's Hospital **Study period**: October 2013 – October 2016. Groups Classification • **Group 1:** (LLF)**:** *n* = 34 • **Group 2:** (RSF): *n* = 60 • **Group 3:** maternal breast milk (MBM)**:** *n* = 16	• **Group 1:** LLF • **Group 2:** RSF feeding, formula • **Group 3:** Infants exclusively fed MBM	• **Duration of MSO_4, treatment, days:** Adjusted *p*-value = .12 **Group 1:** 24 ± 19 **Group 2:** 15 ± 14 • **Length of hospital stay, days (SD):** **Group 1:** 29 ± 19 **Group 2:** 21 ± 12 • **Need for pharmacological therapy, %:** **Group 1:** 85 **Group 2:** 68 • **Growth:** Measured as % of neonates requiring 20 kcal/oz, 22 kcal/oz, 24 kcal/oz to maintain an adequate growth rate of at least 15 mg/kg/day. **Group 1:** 44% required some increased caloric fortification **Group 2:** 27% required some increased caloric fortification	(0.12) (0.16) (0.86) NA
Kaplan et al., 2020 [12]	**Setting:** At 47 sites, The Ohio Perinatal Quality Collaborative (OPQC) conducted a quality improvement initiative using orchestrated testing (OT) to improve non-pharmacological care of infants dealing with NAS requiring pharmacological treatment. **Study period:** October 2015 to June 2016. **Group Classification:**	**Group 1, low lactose formula (LL), high-calorie formula (HC):** Fortify feeds to 22 kcal/oz **Group 2, high lactose formula (HL), high calorie formula (HC):** Fortify feeds to 22 kcal/oz **Group 3, low lactose formula (LL), low calorie formula (LC):** Missing specifics **Group 4, high lactose formula (HL), low calorie formula (LC):** Missing specifics	• **Length of hospital stay** **Base Analysis:** Group 1 (LL+HC): 16 days Group 2 (HL+HC): 18.7 days Group 3 (LL+LC): 18.7 days Group 4 (HL+LC): 18 days **Adjusted analysis (accounting for hospital, breastfeeding, weight loss):** Group 1 (LL+HC): 16.6 days Group 2 (HL+HC): 14.8 days Group 3 (LL+LC): 19.7 days Group 4 (HL+LC): 17.6 days • **Treatment failure (dose escalation, failed wean, and/or requires secondary**	Not reported Not reported

	• **Group 1, low lactose formula (LL), high calorie formula (HC):** *n = 94* • **Group 2, high lactose formula (HL), high calorie formula (HC):** *n = 74* • **Group 3, low lactose formula (LL), low calorie formula (LC):** *n = 333* • **Group 4, high lactose formula (HL), low calorie formula (LC):** *n = 54*	Groups may discontinue specific formula guidelines once infant has been weaned off opiates for 24 hours.	**pharmacological treatment):** **Base Analysis:** Group 1 (LL+HC): 54% Group 2 (HL+HC): 62% Group 3 (LL+LC): 71% Group 4 (HL+LC): 58% **Adjusted analysis (accounting for hospital, breastfeeding, weight loss):** Group 1 (LL+HC): 63% Group 2 (HL+HC): 44% Group 3 (LL+LC): 73% Group 4 (HL+LC): 56% • **Weight loss (>10% within 7 days of life)** **Base Analysis:** Group 1 (LL+HC): 7% Group 2 (HL+HC): 13% Group 3 (LL+LC): 13% Group 4 (HL+LC): 14% **Adjusted analysis (accounting for hospital, breastfeeding, weight loss):** Group 1 (LL+HC): 3% Group 2 (HL+HC): 3% Group 3 (LL+LC): 4% Group 4 (HL+LC): 4% • **Pharmacological intervention(s):** Group 1 (LL+HC): Balanced use of methadone or morphine Group 2 (HL+HC): Balanced use of methadone or morphine Group 3 (LL+LC): Favored use of morphine (85% of sites) Group 4 (HL+LC): Balanced use of methadone or morphine	Not reported Not reported Not reported Not reported Not reported
Lembeck et al., 2020 [13]	**Setting:** The pediatric floor of Einstein Medical Center, USA. **Study period:** July 2014-November 2016	**Group 1,** Majority or exclusively breastfeeding **Group 2,** Majority or exclusively RSF	• **Duration of NAS:** Not specified • **Length of hospital stay, (day) (adjusted for gestational age) Mean (CI)**	

	Group classification: • **Group 1a,** Majority Breastfeeding (MB)**:** *n* = 65 • **Group 1b,** Exclusively Breastfeeding (EB): *n* = 39 • **Group 2a,** Majority RSF (MRSF)**:** *n* = 147 • **Group 2b,** Exclusively RSF (ERSF): *n* = 105 • **Group 3a,** Majority Low-Lactose Formula (MLLF): *n* = 37 • **Group 3b,** Exclusively Low – Lactose Formula (ELLF): *n* = 14	**Group 3,** Majority or exclusively Low-Lactose Formula	**{p-value} (Majority feeding infants)** MLLF vs MRSF: 3.0, MBF vs MLLF: -10.3 MBF vs MRSF: -7.4 **Length of hospital stay, d (adjusted for gestational age) Mean (CI) {p-value} (Exclusive feeding infants)** ELLF vs ERSF: -1.1 EBF vs ELLF: -7.4 EBF vs ERSF: -8.5 **The length of pharmacological intervention, d (adjusted for gestational age), Mean (CI) {p-value} (Majority feeding infants)** MLLF vs MRSF: 3.9 MBF vs MLLF: -10.8 MBF vs MRSF: -6.9 **The length of pharmacological intervention, d (adjusted for gestational age) Mean (CI) {p-value} (Exclusive feeding infants)** ELLF vs ERSF: -0.5 EBF vs ELLF: -7.9 EBF vs ERSF: -8.4 **Growth, Weight change (g/day) (Adjusted for length of stay): Mean (CI) {p-value} (Majority feeding infants)** MLLF vs MRSF: -2 MBF vs MLLF: -9.9 MBF vs MRSF: -12.3 **Growth, Weight change birth to discharge (gestational/discharge) (Adjusted for length of stay): Mean (CI) {p-value} (Exclusive feeding infants)** ELLF vs ERSF: 6 EBF vs ELLF: -24	(.15),{-1.1, 7.0} (< .01),{-14.9, -5.8} (<.01),{-10.7 to-4.1} (.93), {-8.4, 6.3} (.04), {-14.6, -0.3} (< .01),{-12.4, -4.6} (0.08), {-0.4, 8.1} (< .01),{-15.6, -5.9} (< .01){-10.5, -3.4} (0.89),{-7.5, 6.5} (.04),{-15.6, -0.3} (< .01),{-13.1, -3.8} (.62), {-11.7, 6.9} (.07), {-20.8, 1.0} (< .01),{-20.2, -4.3}

			EBF vs ERSF: -18	(0.39),{-8, 20} (< .01){-39, -9} (<.01),{-27, -8}
Pandey et al., 2020 [14]	**Setting:** MetroHealth Medical Center, Cleveland, Ohio **Study Period** December 2014 to June 2018 **Completed Study:** *N* = 69 • **Group 1:** *n* = 34 • **Group 2:** *n* = 35 • **Gender, males (%):** • **Group 1:** *n* = 20 (57%) • **Group 2:** *n* = 18 (51%) **Power Analysis:** Projected sample size of 32 patients per group would allow detection of an effect size of 0.70 (reduction in opiate usage/standard deviation) with a power of 80% and type I error of 0.05 (two-tail). Risk of bias is shown in table 3.	**Group 1:** lactose-free formula (Similac Sensitive® reconstituted from concentrated liquid to 20 calorie/ounce) **Group 2:** lactose-containing formula (Similac Advance® reconstituted from concentrated liquid to 20 calorie/ounce)	• **Cumulative morphine dose (mg/kg) required to treat NAS** Group 1: 7.2±7.5 Group 2: 7.5±7.6 • **Length of medical treatment** Group 1: 16.5±13.6 Group 2: 19.6± 19.5 • **Length of hospital stay** Group 1: 22.4±14.1 Group 2: 27±19.8 Notes: 1.Secondary outcomes and patient characteristics were similar between treatment groups ($p > 0.100$), except maternal heroin abuse was higher in the lactose-containing group ($p = 0.013$). 2.More patients in the lactose-containing formula group required second-line agents for the treatment of NAS, but the difference was not statistically significant for each individual agent.	(>0.1) (>0.1) (>0.1)